

HISSA

HISSA Financial Data Engine Squad

Internship Work Brief, Job Description, Resources & Delivery Plan

Document	Internal Working Document
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Audience	4 Data Analyst Interns + 1 Full Stack Intern + CTO/System Owner
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Version	1.1
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Date	26 August 2026
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Internship End	19 September 2026
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Purpose. Menjadi acuan tunggal untuk keputusan arsitektur, pembagian tanggung jawab, kualitas data, dan implementasi subsystem Financial Data Engine HISSA.

1. Project Mission

Squad ini bertugas membangun subsystem production-oriented yang mengubah filing laporan keuangan IDX menjadi data fundamental HISSA yang terstruktur, tervalidasi, dapat diaudit, dan dapat dipakai oleh aplikasi. Lima intern bekerja sebagai satu squad dengan ownership berbeda—bukan lima project terpisah.

Core outcome IDX filing → raw facts → normalized financial data → validation → fundamental metrics → debt/sharia intelligence → HISSA Ops review → publish.

1.1 Working principles

- Evidence over assumption: setiap mapping/angka harus punya sumber.
- Structured data first: XBRL/instance sebelum XLSX/PDF/AI.
- Do not guess: jika makna/data tidak jelas, tandai UNMAPPED / REVIEW_REQUIRED.
- Reproducible work: script, notebook, mapping, dan test harus dapat dijalankan ulang.
- One squad: temuan Data Analyst harus segera diterjemahkan menjadi contract yang dapat diimplementasikan engineer.
- **Production mindset**: tidak boleh bergantung pada edit SQL manual atau langkah yang hanya diketahui satu orang. Intern development wajib berjalan di sandbox terisolasi dengan prinsip least privilege.

2. Team Structure & Ownership

Role	Primary ownership	Primary deliverable
DA-1 – IDX & XBRL Analyst	Source structure, taxonomy, contexts, entry points	IDX/XBRL Source Map + reproducible raw extraction dataset
DA-2 – Financial Mapping Analyst	Financial semantics and canonical concepts	Canonical Financial Dictionary + mapping rules
DA-3 – Data Quality Analyst	Validation, reconciliation, anomalies	Validation Rulebook + quality report/test cases
DA-4 – Debt & Sharia Analyst	Debt notes, lender data, financing classification	Debt Dictionary + pilot creditor dataset + evidence
Full Stack – Financial Platform Engineer	Pipeline, DB, queue, Vue/Inertia Ops	Working ingestion pipeline + HISSA Ops MVP
CTO – System Owner	Architecture, data contracts, priority, acceptance	Architecture decisions, unblockers, final acceptance

Important Role ownership bukan silo. DA-1/2/3/4 wajib saling review; Full Stack wajib menggunakan contract/data rules yang disepakati. Semua perubahan mapping dan rule harus terdokumentasi.

3. Job Description – DA-1 IDX & XBRL Analyst

- **Mission.** Mengetahui dengan presisi format data yang disediakan IDX dan cara mengambil fakta yang benar dari filing.
- **Research.** Pelajari instance.zip, inlineXBRL.zip, XLSX, PDF, taxonomy, entry point, context, unit, decimals, dimensions.
- **Sampling.** Ambil filing pilot lintas industri dan periode; dokumentasikan variasi struktur.

- **Extraction prototype.** Buat script/notebook yang membaca XBRL dan menghasilkan daftar fact/context/unit tanpa hard-code ticker tertentu.
- **Context rules.** Identifikasi consolidated vs parent, current vs comparative, instant vs duration, current period vs prior period.
- **Taxonomy inventory.** Catat concepts/notes penting termasuk bank loans, bonds, sukuk, borrowings.

Required outputs

Output	Minimum acceptance
idx-taxonomy-notes.md	Entry points, contexts, units, period conventions, useful note nodes.
pilot_raw_facts.csv/parquet	Must-have: 10 issuers × minimum 4 periods (≥40 filings). Stretch: 20 issuers × up to 8 periods; source identity retained.
extract_xbri prototype	Re-runnable; produces facts + contexts + units + dimensions; no manual copy-paste.
source-issues.md	Known source anomalies, missing data, revisions, inconsistent patterns.

Definition of done Orang lain dapat mengambil satu filing pilot, menjalankan extractor, dan memperoleh output yang sama tanpa bantuan DA-1.

4. Job Description – DA-2 Financial Mapping Analyst

- **Mission.** Membangun kamus finansial HISSA sehingga konsep yang berbeda dapat dibandingkan secara konsisten.
- **Canonical dictionary.** Definisikan 30–50 konsep awal untuk Income Statement, Balance Sheet, dan Cash Flow.
- **Semantic mapping.** Map source concept IDX → canonical concept berdasarkan makna, bukan kemiripan nama saja.
- **Metadata.** Tentukan statement, instant/duration, sign convention, allowed scope, required/optional, formula dependency.
- **Industry exceptions.** Dokumentasikan konsep yang tidak comparable lintas entry point dan perlakuannya.
- **Review.** Peer-review mapping bersama DA-1 dan DA-3.

Output	Minimum acceptance
canonical_financial_dictionary.csv	Minimum 30 core concepts for Income Statement, Balance Sheet, and Cash Flow; stretch toward 50 after pilot stability.
concept_mapping.csv	source_concept, entry_point, canonical_concept, status, rationale, reviewer.
mapping-rules.md	Rules untuk consolidated scope, sign, period, and known exceptions.
mapping test pack	Samples yang membuktikan mapping menghasilkan nilai yang sama dengan filing sumber.

Definition of done Minimal 30 core concepts mempunyai definisi yang tidak ambigu dan mapping pilot yang dapat direview serta diimplementasikan engineer.

5. Job Description – DA-3 Data Quality & Reconciliation Analyst

- **Mission.** Mencegah data salah masuk production.
- **Accounting rules.** Bangun checks seperti Assets $\approx$ Liabilities + Equity dan hierarchy checks.
- **Context checks.** Detect wrong period, wrong scope, duration/instant mismatch, duplicate facts.
- **Scale checks.** Detect multiplier/currency/unit anomalies dan magnitude yang tidak masuk akal.
- **Cross-period.** Detect revisions/restatements dan comparative inconsistencies.
- **Quality scoring.** Definisikan VERIFIED / REVIEW_REQUIRED / FAILED dan severity ERROR/WARN/INFO.
- **Test cases.** Sengaja buat contoh data rusak dan pastikan rule mendeteksinya.

Output	Minimum acceptance
validation-rulebook.md	Rule code, description, severity, inputs, tolerance, expected behavior.
validation_cases.csv	Positive and negative test cases with expected result.
pilot-quality-report.md	Quality score/status per filing, unresolved anomalies, recommendation.
reconciliation checklist	Manual verification checklist against IDX/issuer source.

Definition of done Sebuah mapping/value yang salah secara sengaja dapat ditangkap oleh rule atau masuk REVIEW_REQUIRED sebelum publish.

6. Job Description – DA-4 Debt & Sharia Intelligence Analyst

- **Mission.** Membuktikan bahwa HISSA dapat mengubah disclosure utang menjadi creditor/debt intelligence yang memiliki bukti.
- **Structured-first.** Inventaris node XBRL untuk short/long-term bank loans, borrowings, bonds, sukuk, interest/sharia income/expense.
- **Debt model.** Definisikan borrower, creditor, facility, outstanding, limit, currency, rate, maturity, collateral, purpose.
- **Normalization.** Buat creditor normalization rules dan debt category dictionary.
- **Classification.** CONVENTIONAL / ISLAMIC / MIXED / UNDETERMINED; sharia contract hanya jika ada evidence.
- **Coverage.** Hitung identified vs aggregate debt; jangan menyatakan 100% jika lender tidak teridentifikasi.
- **Evidence.** Simpan source document, section/page, text/table reference, extraction method.

Output	Minimum acceptance
debt_dictionary.csv	Field definitions, allowed categories, null/unknown behavior.
creditor_master.csv	Normalized creditor names, type, country where known; alias handling.
pilot_debt_dataset.csv	Must-have: at least 5 pilot issuers with aggregate debt + lender/facility details where disclosed. Stretch: 10 issuers.
debt-evidence pack	Every material lender/facility record points to source evidence.
sharia-classification-notes.md	Evidence-based classification rules and unresolved cases.

Definition of done Pilot dapat menampilkan creditor coverage %, unidentified amount, dan evidence; tidak ada classification yang hanya menebak dari nama bank.

7. Job Description – Full Stack Financial Platform Engineer

- **Mission.** Mengubah research/data rules menjadi subsystem yang otomatis, dapat dioperasikan, dan dapat diaudit.
- **Backend.** Laravel models/services/jobs di dedicated sandbox repository; ingestion, normalized store, validation execution, publish workflow sandbox.
- **Queue.** Redis + Horizon; separate queues for downloads, XBRL, normalize, validate, analytics, enrichment, publish.
- **Python worker.** Integrasi Arelle parser; clean JSON/DB contract; process isolation untuk parsing.
- **Storage.** Simpan file source dengan stable path + hash; raw facts immutable.
- **Admin.** Build HISSA Ops sandbox dengan Vue 3 + Inertia: Pipeline, Financial Review, Concept Mapping, Data Quality, dan Debt Review jika core selesai.
- **Isolation & integration.** Intern tidak diberi akses HISSA Core source code, production database, production secrets, atau production infrastructure. Deliver subsystem melalui stable API/JSON/data contract; integrasi ke HISSA Core dilakukan CTO/core engineering.
- **Testing.** Automated tests for core data flow and critical Ops actions.

Output	Minimum acceptance
Database migrations	Raw, normalized, validation, metrics, debt, audit structures.
Pipeline jobs	Discovery → download → parse → normalize → validate → publish; retries/idempotency.
Python/Arelle worker	Parses pilot XBRL and returns structured facts consistently.
HISSA Ops MVP	Pipeline status, review, mapping, validation, retry/reprocess, audit actions.
Runbook	How to deploy workers, start Horizon, reprocess a filing, diagnose failures.
Sandbox repository	Dedicated private project runs independently from HISSA Core; reproducible setup, sample data, no production dependency.

Definition of done Seorang Data Analyst dapat menemukan filing bermasalah, melihat source/evidence, mengoreksi mapping melalui workflow yang disediakan, lalu reprocess tanpa edit database manual.

8. Shared Data Contract

Semua anggota squad wajib menggunakan contract yang sama. Contoh normalized fact:

```
{
  "ticker": "ANTM",
  "period": "2026-Q2",
  "canonical_concept": "total_assets",
  "value": 42800000000000,
  "currency": "IDR",
  "scope": "consolidated",
  "data_type": "reported",
  "source_concept": "...",
  "filing_id": "...",
  "validation_status": "verified"
}
```

- Jika field belum diketahui, gunakan null/status explicit—jangan invent data.
- Semua perubahan contract harus diputuskan bersama Full Stack + relevant DA + CTO. Contract harus cukup stabil agar HISSA Core dapat mengonsumsi output tanpa membuka source code atau database production kepada intern.
- Mapping/rules harus versioned; source concept dan rationale tidak boleh hilang.

9. Pilot Dataset

Must-have pilot: 10 issuer lintas karakter × minimum 4 periods (≥40 filings). Stretch target: 20 issuer × hingga 8 periods hanya jika pipeline inti sudah stabil. Daftar awal dapat diubah setelah DA-1 mengonfirmasi entry-point coverage.

Category	Suggested examples	Reason
Bank / Financial	BBCA, BRIS	Different financial/sharia reporting structure.
Mining / Resources	ANTM, AADI	Debt/CALK complexity and commodity businesses.
Telco / Infrastructure	TLKM	Large consolidated issuer and capex/debt.
Conglomerate / Heavy industry	ASII, UNTR	Multiple businesses/subsidiaries.
Manufacturing	SMGR, CPIN	Inventory, COGS, operating accounts.
Property	PWON or peer	Property entry point variation.
Others	Add until 10 must-have; stretch 20	Ensure diverse contexts and taxonomy behavior.

Sampling rule Jangan memilih hanya perusahaan yang datanya “rapi”. Pilot harus sengaja mencakup variasi dan edge cases.

10. Accelerated Delivery Sprint — 26 Aug to 19 Sep 2026

Timebox: 26 August 2026 sampai 19 September 2026 (25 calendar days). Semua role bekerja paralel; blocker yang menghambat dependency wajib dieskalasikan pada hari yang sama.

Windo w	Squad objective	Mandatory output / exit gate
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26–28 Aug	Sandbox bootstrap + source proof	Private sandbox repo, dev environment, approved sample filings, first reproducible XBRL extraction, task board, shared contract v0.
29 Aug–4 Sep	Financial core MVP	Source/context inventory, ≥30 canonical concepts, mapping draft, raw + normalized schema, parser contract, first normalized pilot dataset.
5–10 Sep	Validation + automation MVP	Period/scope rules, validation rulebook, queue/jobs, idempotency/revision handling, ≥40 pilot filings process through raw → normalized → validation.
11–14 Sep	HISSA Ops + intelligence pilot	Vue/Inertia Pipeline + Financial Review + Data Quality, fundamental metric proof, debt/sharia pilot ≥5 issuers with evidence and coverage.
15–18 Sep	Hardening + handover freeze	Fix blocking defects, regression tests, documentation, API/JSON contract, runbook, architecture notes, final quality report, demo rehearsal. No new scope after 17 Sep.
19 Sep	Final demo + handover	End-to-end demo on non-hand-cleaned filing, repository handover, known limitations/backlog, final acceptance package for CTO/core team.

Scope freeze: internship tidak menargetkan full IDX backfill, production deployment, direct write ke HISSA database, atau rewrite seluruh Filament. Fokus wajib: pilot end-to-end yang reproducible, tervalidasi, dan siap handover.

11. Accelerated Working Rhythm

Cadence	Activity	Expected result
Daily start	10–15 minute standup: done / today / blocker / dependency	Every person has one measurable output for the day; blockers become visible immediately.
Daily	Async evidence update in issue/task	Commit, dataset, screenshot, test result, or source evidence is attached; no status-only updates.
Same day	Data/engineering unblock when contract or semantics block another role	No waiting until weekly meeting for mapping/context/schema questions.
2× per week	Integrated demo + quality review	Show running pipeline/output on real pilot filings, not only slides.
2× per week / as needed	CTO decision review	Resolve architecture, financial semantics, scope, and security boundary decisions quickly.

- Every ticket must have owner, input, expected output, acceptance criteria, and evidence/source link.
- Research findings that affect software must become an issue/ADR/rule—not remain only in chat.
- Pull requests require concise description, test evidence, and reference to the task/issue.

12. Definition of Done

1. Output is reproducible from repository/scripts, not manual copy-paste.
2. Source/evidence is recorded and accessible.
3. Assumptions and unresolved ambiguity are documented.
4. Relevant unit/integration/data validation tests pass.
5. Peer review has occurred for mapping or financial semantics changes.
6. No blocking validation errors remain for data proposed as VERIFIED.
7. Documentation/runbook updated when behavior or setup changes.
8. Demo proves the result works on more than one hand-picked filing.

13. Tools & Recommended Stack

Area	Recommended tool	Usage
Source research	IDX website + browser	Locate filings, taxonomy docs, verify source.
XBRL	Arelle + Python	Parse/validate instance/iXBRL; inspect contexts/units/dimensions.
Data analysis	Python, pandas, Jupyter optional	Exploration, reconciliation, sampling; production rules must migrate to maintainable code.
Backend	Laravel 12	Jobs, services, validation orchestration, APIs/controllers.
Queue	Redis + Laravel Horizon	Async workload and observability.
Database	MySQL	Raw metadata, normalized facts, rules, metrics, audit.
Admin frontend	Vue 3 + Inertia	HISSA Ops.
Version control	Dedicated private Git repository	Intern squad works only in Financial Data Engine sandbox repo; branches/PRs/issues; HISSA Core repository is out of scope.
API testing	Postman/Bruno/curl	Internal endpoint checks.

14. Sandbox Repository, Access Policy & Environment Setup

- Clone hanya dedicated private Financial Data Engine sandbox repository dan confirm branch/PR workflow. Jangan clone/copy HISSA Core repository.
- Install PHP/Composer dan dependencies berdasarkan sandbox repository/lockfile. Sandbox harus dapat berjalan tanpa source code HISSA Core.
- Install Node/npm compatible dengan sandbox; Vue/Inertia dependencies wajib melalui repository lockfile.
- Gunakan MySQL dan Redis lokal/sandbox atau approved Docker/dev services. Dilarang memakai production database sebagai development environment.
- Install Python virtual environment dan pinned parser dependencies; install/test Arelle pada sandbox worker.
- Gunakan hanya minimum sandbox credentials yang disediakan. Production .env, API keys, database credentials, SSH keys, dan infrastructure credentials tidak diberikan kepada intern.
- Run seed/sample pipeline pada satu approved filing. Integrasi ke HISSA Core hanya melalui approved API/JSON/data contract dan dilakukan CTO/core engineering.

15. What You Must Not Do

Do not	Why
Guess concept mapping or sharia classification	Incorrect semantics are worse than missing data.
Use LLM output as final financial value without source reconciliation	Hallucination / extraction error risk.
Edit production database manually to "fix" pipeline output	Breaks reproducibility and auditability.
Overwrite old filing values on revision	Destroys restatement history.
Hard-code only for one ticker without documenting why	Will fail when scaling.
Download/redistribute data externally without permission	Data rights and licensing must be respected.
Rewrite all Filament pages before Ops MVP works	Creates product risk without improving data pipeline.
Clone/copy/access HISSA Core source code or production repository	Internship work is sandbox-only; protect proprietary source and enforce least privilege.
Connect sandbox directly to HISSA production DB or write production data	Integration is contract-based and executed by CTO/core engineering after review.
Use production credentials, secrets, or infrastructure for development	A sandbox compromise or bug must not expose production systems.

16. Escalation Rules

Situation	Action
Financial meaning ambiguous	DA-2 proposes options + source evidence → peer review → CTO decision if material.
XBRL context/technical ambiguity	DA-1 + Full Stack reproduce case; document minimal fixture.
Validation false positive/negative	DA-3 updates rule/test proposal; do not bypass silently.
Debt/sharia classification uncertain	DA-4 marks UNDETERMINED and records evidence/question.
Architecture/schema change	Open issue/ADR; Full Stack + relevant DA + CTO review before migration.
Licensing/redistribution question	Escalate to CTO/business/legal owner; do not assume.

17. Internship Evaluation Rubric

Dimension	Weight	What good looks like
Technical/data correctness	30%	Source-backed, accurate, handles edge cases.
Reproducibility & engineering quality	20%	Scripts/tests/docs; no manual-only workflow.
Problem solving	15%	Finds root cause and proposes bounded solutions.
Collaboration	15%	Cross-role review, clear handoffs/contracts, unblocks team.
Ownership	10%	Tracks deliverables, communicates blockers early.
Documentation/demo	10%	Clear evidence, runbook, understandable final demo.

18. Final Demo Checklist

- Select a filing pilot that has not been manually pre-cleaned.
- Show discovery/download and stored original artifact.
- Show extracted XBRL facts and source context.
- Show canonical normalized financial values.
- Show validation results and one intentionally caught issue/edge case.
- Show fundamental metric derived from stored inputs.
- Show one debt-intelligence example with creditor coverage and evidence.
- Show HISSA Ops flow: review → mapping/approval → reprocess → publish.
- Show revision/restatement handling or a reproducible test demonstrating it.
- Present handover package: sandbox repository, setup/runbook, stable API/JSON contract, limitations, unmapped concepts, scale risks, licensing decision, and recommended HISSA Core integration steps.

Reference Resources

Gunakan dokumentasi resmi sebagai sumber utama. Tanggal akses: 26 August 2026.

R1 IDX – Financial Statements & Annual Report

<https://www.idx.id/en/listed-companies/financial-statements-and-annual-report>

Sumber filing publik; contoh 2026 tersedia PDF, XLSX, instance.zip, dan inlineXBRL.zip.

R2 IDX – XBRL & IDX Taxonomy

<https://www.idx.id/en/listed-companies/xbrl/>

Penjelasan XBRL IDX, entry points, taxonomy, notes expansion, dan materi pendukung.

R3 IDX – Data Services / Data Reference

<https://www.idx.id/en/products/idx-data-services/>

Acuan untuk evaluasi lisensi display/non-display/redistribution sebelum production commercial use.

R4 Laravel 12 – Queues

<https://laravel.com/docs/12.x/queues>

Queue jobs, retry, batching, prioritization, and error handling.

R5 Laravel 12 – Horizon

<https://laravel.com/docs/12.x/horizon>

Monitoring dan konfigurasi Redis queues.

R6 Laravel 12 – Task Scheduling

<https://laravel.com/docs/12.x/scheduling>

Scheduler untuk filing discovery dan housekeeping.

R7 Inertia.js – Official

<https://inertiajs.com/>

Modern monolith: Vue pages dengan routing/controller Laravel, tanpa API terpisah untuk admin internal.

R8 Vue 3 – Guide

<https://vuejs.org/guide/introduction.html>

Vue 3, Composition API, Single-File Components.

R9 Arelle – Open Source XBRL Platform

<https://arelle.org/arelle/>

Parser/validator XBRL dengan CLI, web service, dan Python API.

R10 XBRL International – XBRL 2.1

<https://specifications.xbrl.org/work-product-index-group-base-spec-base-spec.html>

Referensi standar XBRL 2.1.

R11 XBRL International – iXBRL Overview

<https://www.xbrl.org/the-standard/what/ixbrl/>

Konsep Inline XBRL: human-readable + machine-readable.

Questions / architecture decisions: escalate through the squad issue tracker and CTO review. Do not leave critical decisions only in private chat.